

Habit, Not Infrastructure: Motorcycle Lock-In and Micro-Enterprise Vulnerability as a Behavioural Baseline on the Eve of Metro Line 1 Operation in Ho Chi Minh City

Highlights

- Behavioural attribution outweighs infrastructural attribution 6.1:1 (68% vs 11.1%, $n = 460$).
- Residents who cite infrastructure do not live further from stops than those who cite habit.
- The 71 residents who call walking to stops uncomfortable have statistically identical walks to everyone else.
- Motorcycle dependence rises with household income (OR 1.76 per band), ruling out affordability.
- Officials attribute non-use to travel time (57%); residents attribute it to habit (70%).

Abstract

Ho Chi Minh City opened Metro Line 1 on 22 December 2024, through a corridor where private motorcycles carried 83.5% of trips and buses carried 1.7%. This paper reports the behavioural baseline recorded in that corridor immediately before the line began commercial service, and tests whether the barrier structure it reveals is the one that infrastructure investment can dissolve. Face-to-face surveys of 460 residents, 80 micro-enterprises and 23 local officials in the Truong Tho ward cluster of Thu Duc City are combined with two dated OpenStreetMap snapshots, four global settlement-layer rasters and two Sentinel-2 built-up indices, giving an objective accessibility measurement for every surveyed dwelling at two points in time. Residents attribute non-use of public transport to habit 68% of the time and to infrastructure deficits 11.1% of the time, a ratio of about 6.1:1. Crucially, the stated reasons do not correspond to measured conditions: network walking distance, straight-line distance and the pedestrian detour ratio are statistically indistinguishable across the barrier groups (all $p \geq 0.075$, $\varepsilon^2 \leq 0.010$), and the 71 residents who report that walking to stops is uncomfortable have walks that are no longer, no more circuitous and no less well connected than everyone else's ($p = 0.73$ to 0.98). Reclassifying that group as perceptual rather than physical raises the ratio to about 7.5:1. A logistic model of motorcycle dependence finds no accessibility predictor significant while household income is positive and significant (OR 1.76, 95% CI 1.19–2.62, $p = 0.005$); three cross-validated classifiers reach ROC-AUC of only 0.61–0.64 and none beats the majority-class rate, indicating a mode choice that is near-universal rather than conditioned on circumstance. Local officials, answering the same question, attribute non-use overwhelmingly to travel time (12 of 21) rather than to habit, a 51-percentage-point diagnostic gap ($\chi^2(3) = 129.56$, $p < 0.001$). Micro-enterprises in the corridor are almost entirely B2C kerbside businesses whose customer flow is organised around the ability to stop a motorcycle at the door. We argue that the motorcycle in this setting is not a

vehicle but the load-bearing infrastructure of a street-level economy, and that mode-shift policy which addresses only the supply side will neither meet its environmental targets nor protect the livelihoods it disturbs.

Keywords: travel habit; socio-technical lock-in; motorcycle dependence; metro; micro-enterprise; OpenStreetMap; Ho Chi Minh City

1 Introduction

On 22 December 2024, after nearly two decades of construction, Ho Chi Minh City's first metro line entered commercial service. Line 1 runs 19.7 km from Ben Thanh in the historic centre to the Suoi Tien terminal in the north-east, and for most of that distance it follows the Hanoi Highway through Thu Duc City. The Truong Tho ward cluster sits directly on this alignment. Between 2022 and 2023, in the window immediately preceding commercial operation, the Truong Tho Living Lab conducted face-to-face surveys of 460 residents, 80 micro-enterprises and 23 local officials in that cluster. The resulting dataset records how people moved, and why they did not use public transport, in the last period in which that question could be asked without the metro already forming part of the answer. It cannot be collected again.

What the baseline records is a corridor at the far end of motorcycle dependence. 384 of 460 respondents (83.5%) name a private motorcycle as their main mode; 1.7% name the bus. The interesting finding, however, is not the modal split but the structure of the explanation people give for it. Asked why they do not use public transport, 68% of respondents choose the habit of using a motorcycle. Only 11.1% choose either of the two infrastructural options offered — insufficient stop density or frequency, and the absence of parking for a personal vehicle at stops. The ratio is about 6.1 to one.

Taken at face value, a ratio of that size is awkward for the dominant policy logic. Vietnamese urban transport strategy, like that of most rapidly motorising cities, treats mode shift as a supply problem: build the rail, add the feeder buses, and demand will follow. If the binding constraint is instead an automated behavioural routine, then supply-side investment addresses roughly one ninth of the stated problem. But a stated reason is not evidence. Respondents may rationalise; "habit" may be a socially easy answer that conceals a real infrastructural grievance; and self-reported barriers are precisely the class of data that reviewers of behavioural transport research are trained to distrust.

This paper's methodological contribution is to subject those stated reasons to an external test. Because OpenStreetMap is archived as dated national extracts, it is possible to reconstruct the mapped public-transport network of Viet Nam as it stood on 1 January 2023 — inside the survey window — and again on 1 January 2026, one year after Line 1 opened, and to compute for every surveyed dwelling the straight-line distance, the walking distance along the pedestrian network, and the ratio between them, at both dates. The question then becomes empirical rather than interpretive: do the people who say the problem is infrastructure actually face worse infrastructure than the people who say the problem is habit?

They do not. Across five stated-reason groups, none of the three accessibility measures differs significantly, and the effect sizes are negligible ($\varepsilon^2 \leq 0.010$). The sharpest case is the group of 71 respondents (15.4%) who say they are uncomfortable walking to stops — the one answer that sits ambiguously between a physical condition and a preference. Their median network walking distance

is 450 m against 398 m for everyone else, their detour ratio is identical to two decimal places, and neither difference approaches significance. The barrier they describe is not in the street network. Reclassifying them accordingly raises the behavioural share to 83.4% against an unchanged 11.1% infrastructural share.

Three further findings sharpen the argument. First, a logistic model of motorcycle dependence finds household income positively and significantly associated with motorcycle use (OR 1.76 per income band, $p = 0.005$) while every accessibility variable is null — motorcycle dependence in this corridor is not a constraint of poverty that rising incomes will relieve, and three cross-validated machine-learning classifiers confirm that the outcome is barely predictable from circumstance at all. Second, the local officials responsible for the corridor, answering the same question about the same population, attribute non-use overwhelmingly to travel time rather than to habit; the diagnostic gap between the governed and the governing is 51 percentage points. Third, the corridor’s micro-enterprises — 75 of 80 with five staff or fewer, 70 of 80 selling directly to consumers — depend on exactly the kerbside stopping behaviour that a successful mode shift would remove.

We therefore frame the motorcycle in this setting not as a vehicle but as the load-bearing infrastructure of a street-level economy: shopfront retail, last-mile delivery and drop-in custom are all organised around the ability to stop a two-wheeler at a door. A transition away from it is not a change of travel mode but a reorganisation of livelihoods. The paper’s four research questions follow directly:

RQ1. What is the quantitative structure of barriers to public-transport use in a motorcycle-dependent corridor immediately before a metro line opens?

RQ2. Do self-reported barriers correspond to objectively measured accessibility conditions, and what predicts motorcycle dependence?

RQ3. How exposed are corridor micro-enterprises to a restructuring of passing trade, and does proximity to a future station shape their outlook?

RQ4. Do local officials diagnose the barrier in the same terms as the residents they govern?

2 Conceptual framework

2.1 Habit and automaticity in travel behaviour

The first pillar is the treatment of mode choice as a habit rather than a decision. Verplanken and Aarts (1999) characterised habit as goal-directed automaticity: a behaviour repeated in a stable context ceases to be re-evaluated and is triggered directly by situational cues. Verplanken et al. (1998) showed experimentally that respondents with strong travel habits acquired less information before choosing, and Aarts et al. (1998) demonstrated that past behaviour predicts future travel mode over and above intention. Bamberg et al. (2003) integrated the finding into the theory of planned behaviour, and Gardner (2009) confirmed that in stable contexts habit outperforms motivation in predicting mode.

The policy implication is uncomfortable: an automated behaviour is by construction insensitive

to new information about alternatives. Fujii and Kitamura (2003) found that a one-month free bus pass changed habitual drivers' behaviour only while the intervention interrupted the routine. Schwanen et al. (2012) argue that habit should not be treated as a residual error term in choice models but as a distinct object requiring its own policy instruments. Thøgersen (2006) reaches a similar conclusion from panel data. A corridor in which 68% of residents name habit as their reason for non-use is, in this literature's terms, a corridor in which supply-side improvement has a weak transmission mechanism to behaviour.

Recent accessibility research strengthens the need to separate measured opportunity from experienced suitability. Ma and Cui's (2024) review treats perceived accessibility as a construct shaped jointly by objective conditions, individual capacities, service quality and preferences. Negm and El-Geneidy (2025), using both calculated and perceived measures, find that the two are only weakly spatially aligned and that perceived accessibility contributes independently to travel behaviour after objective accessibility is controlled. This distinction is central here: an objectively short walk need not be experienced as a usable public-transport journey, while a reported barrier need not correspond to greater metric distance.

2.2 Socio-technical lock-in and path dependence

The second pillar raises the finding from individual psychology to system dynamics. Arthur (1989) and David (1985) established that increasing returns to adoption can lock a society into a technological trajectory that is not the best available. Unruh (2000, 2002) extended the logic to energy systems as carbon lock-in, in which technology, infrastructure, organisations and institutions co-evolve into a mutually reinforcing regime. Seto et al. (2016) distinguish infrastructural, institutional and behavioural lock-in and note that the three operate on different time constants.

Applied to Ho Chi Minh City, the motorcycle is not merely a popular vehicle: it is the centre of a complete socio-technical regime comprising fuel and repair networks at the scale of the alley, informal parking on every pavement, a shopfront retail geometry calibrated to kerbside stopping, and a last-mile delivery labour market. Truitt (2008) documented how motorbike mobility became constitutive of urban middle-class identity in the city; Hansen (2017) traces the parallel emergence of automobility in Hanoi. Lock-in of this kind is not dissolved by adding a rail line; the rail line enters as a competitor to a regime that supplies far more than transport.

Recent Vietnamese evidence reinforces that interpretation without reducing motorcycle use to a single cause. Le et al. (2024) frame Viet Nam's motorcycle dominance as the outcome of interacting psychological, practical and contextual advantages rather than price alone. Kieu et al. (2024), analysing public responses to a proposed Hanoi motorbike ban, further show why restriction policies must be interpreted through distributional and behavioural responses rather than vehicle counts alone. The emerging 2026 research agenda now includes both adoption of electric motorcycle ride-hailing and the response of motorcycle-dependent older adults to fare-free public transport (Nguyen et al., 2026a, 2026b). These developments position motorcycle dependence as a socially embedded mobility system that may be electrified, priced or regulated without necessarily being displaced.

2.3 The multi-level perspective

Geels (2002, 2012) frames transitions as the interaction of niche innovations, an incumbent regime and a landscape of slow exogenous pressures. Metro Line 1 is best read as a regime-level intervention that arrives from above rather than a niche that grows from below, and the 2022–2023 baseline records the regime exactly as it stood before the shock. Köhler et al. (2019) note that transitions research has under-studied the distributional consequences of regime change; the micro-enterprise strand of this paper is a direct response to that gap.

The transferability of conventional transit-oriented development is therefore an empirical question. Pojani and Stead (2015) caution that sustainable-transport instruments developed in wealthier settings cannot be transferred mechanically to rapidly changing cities in the Global South. Vale (2015) likewise shows that station proximity alone is an incomplete TOD measure unless pedestrian network integration is considered. Recent World Bank guidance for emerging cities extends this point by treating TOD as a coordinated package of accessibility, land-use, institutional and inclusion measures rather than rail investment by itself (World Bank, 2024). These arguments motivate the paper’s separation of station distance, pedestrian-network access and livelihood exposure.

2.4 Micro-enterprise dependence on kerbside access

The fourth pillar connects the transport question to the economic one. Kim (2012, 2015) shows that in Ho Chi Minh City the sidewalk is a productive space whose informal property regime is organised around vehicular access, and Turner and Schoenberger (2012) document the fragility of vendor livelihoods in Hanoi when that access is administratively reorganised. Ewing and Cervero (2010) and Cervero and Kockelman (1997) establish that built form and travel interact, but the transit-oriented development literature has generally assumed that increased footfall benefits local retail. That assumption presumes a pedestrian-scaled retail form. Where retail is calibrated to a motorcycle stopping at the door, the substitution of walking passengers for stopping riders is not obviously a gain.

2.5 Integrating proposition

Taken together, the four pillars yield the proposition this paper tests: in a corridor of this type the binding constraint on mode shift is behavioural and systemic rather than infrastructural, and because the incumbent regime is economically as well as physically load-bearing, a mode-shift policy that ignores the micro-economic dimension will fail twice — missing its environmental target and damaging the livelihoods it disturbs.

3 Study context and data

3.1 The corridor

The study area is the Truong Tho ward cluster of Thu Duc City, Ho Chi Minh City, together with six adjoining wards from which respondents were also drawn (Linh Dong, Binh Tho, Linh Trung, Linh Tay, Phuoc Long A and Linh Chieu). The cluster lies on the Hanoi Highway between the Binh Thai and Thu Duc stations of Metro Line 1. At the time of survey the line was structurally complete and undergoing trial running; commercial service began on 22 December 2024.

3.2 Survey instruments

Three instruments were administered face to face by trained enumerators. The resident questionnaire ($n = 460$) covers household composition, occupation, education, income band, main travel mode, commute time, the reason for not using public transport, and a battery of environmental-burden and neighbourhood-satisfaction items; every record carries a GPS coordinate. The micro-enterprise questionnaire ($n = 80$) covers sector, staff count, trading history, customer model and volume, self-assessed growth and profit, and a five-item Likert battery on competitive advantage, intention to remain, perceived change in the area, SME support policy and legal-administrative support. The officials questionnaire ($n = 23$) covers the same substantive domains from the administrative side, including — critically for this paper — an item asking officials why they believe residents do not use buses, phrased to mirror the residents' own item.

3.3 Open data

Five open sources were used. (i) Two Geofabrik OpenStreetMap country extracts for Viet Nam, dated 1 January 2023 and 1 January 2026, supply transit stops, the road and pedestrian network, and commercial points of interest at two dates. (ii) The Global Human Settlement Layer population grids GHS-POP R2023A for epochs 2020 and 2025, at 100 m. (iii) The GHSL built-up surface grids GHS-BUILT-S R2023A for the same epochs. (iv) Sentinel-2 derived normalised difference built-up indices for 2023 and 2026 and their difference. (v) The Provincial Competitiveness Index published annually by VCCI, 2021 to 2025, used as an external benchmark for the enterprise sample. No source requires registration and all are redistributable, so the analysis is fully reproducible.

3.4 Declared data faults

Four data problems are declared here rather than buried, because two of them affect how the headline numbers should be read.

- *Q2.8 contains a non-reason.* The list of reasons for not using public transport includes the option "I do use public transport", selected by 14 respondents. This is a questionnaire-design fault. Percentages are therefore reported on both denominators: 460 (all respondents, matching earlier Truong Tho Living Lab outputs) and 446 (non-users only). The behavioural share is

68% and 70.2% respectively; the infrastructural share is 11.1% and 11.4%. No conclusion in this paper depends on the choice.

- *Q2.8 is single-select.* Respondents could name only one reason, so the instrument measures the dominant perceived barrier rather than the full set. We treat this as a limitation of scope, not of validity: the convergence of two thirds of the sample on a single option is itself informative about dominance.
- *Q1.10 is multi-select and stored as a delimited string.* It is parsed by substring; 384 respondents (83.5%) include a private motorcycle. Twenty-one open-text "other" responses were reviewed and five were recoded to ride-hail.
- *One coordinate is unusable.* A single resident record has a transposed longitude and is excluded from spatial analysis only ($n = 459$ for all accessibility measures, $n = 460$ for all survey-only measures).

4 Methods

4.1 Barrier decomposition

The six response options to Q2.8 are grouped into four analytic classes: behavioural (motorcycle habit), experiential (discomfort walking to stops), infrastructural (stop density or frequency; absence of vehicle parking at stops) and service (bus quality and staff conduct). Shares are reported with 95% Wilson intervals, which behave correctly for proportions near a boundary. Independence of the barrier class from household income, education, age tercile, sex and commute time is tested with χ^2 and reported with Cramér's V .

4.2 Accessibility measurement from dated OSM snapshots

Each national extract is processed in two passes over the raw protocol-buffer file. The first pass retains the coordinates of every node inside the study window (106.66–106.85 °E, 10.74–10.92 °N) and records tagged transit, rail and commercial objects; the second pass rebuilds way geometries from that coordinate cache. Splitting the passes into two handler classes is not cosmetic: a single handler applied twice executes every callback on both passes and silently doubles every count.

Six variables are derived per respondent and per snapshot. Straight-line distance to the nearest mapped public-transport stop is computed in UTM zone 48N by k-d tree. Network walking distance is computed on a pedestrian graph built from all walkable highway classes, using a single multi-source Dijkstra pass: a virtual source is attached to the nearest graph node of every stop, so one traversal yields the distance from every graph node to its nearest stop. The workflow follows the graph-theoretic treatment of OpenStreetMap street networks formalised by Boeing (2017) and updated for contemporary OSM network and amenity analysis by Boeing (2025). The detour ratio is the quotient of network and straight-line distance, and is the measure that separates far from awkward to reach — two dwellings 300 m from the same stop may have very different walks if one must negotiate several dead ends. Commercial point-of-interest counts within 1 km and road-centreline density within 500 m provide land-use controls, and distance to the nearest Metro Line

1 station defines the treatment gradient.

4.3 Snapshot comparability audit

OpenStreetMap completeness in Viet Nam increases over time (Neis and Zipf, 2012; Barrington-Leigh and Millard-Ball, 2017), so a stop that appears in the 2026 extract may be newly built or merely newly drawn. Rather than assume, we audit. Every Metro Line 1 station is matched between snapshots by name and position; the growth in each class of mapped object is tabulated; and the composition of the transit-object growth is inspected by tag. The audit determines which comparisons the data can bear, and Section 5.2 reports a result that reshapes the analysis.

4.4 Modelling motorcycle dependence

A binary logistic regression models the use of a private motorcycle as main mode on twelve predictors: income band, education level, age, sex, own-account business ownership, commute-time band, and the six spatial variables measured at the survey date. Continuous predictors are rescaled to interpretable units (per 100 m, per 0.1 of detour ratio, per 10 points of interest, per 1000 m/km²) so that odds ratios are readable. Three classifiers — the same logistic specification, a random forest and a gradient-boosting machine — are then compared under identical stratified 5-fold cross-validation, scored by ROC-AUC, Brier score and accuracy against the majority-class rate, with permutation importance computed on the best. This design responds to methodological reviews showing that mode-choice ML studies are vulnerable to inconsistent validation, inappropriate accuracy metrics and weak behavioural interpretation (Hillel et al., 2021). Martín-Baos et al. (2023) further demonstrate that superior individual-level prediction does not guarantee credible behavioural indicators or aggregate mode shares, while Dahmen et al. (2024) illustrate the value of interpretable ML for revealed-preference mode data. The machine-learning layer is therefore a robustness check on the linear specification, not a competing causal model: with 83.5% of the sample on one side of the outcome, discrimination is more informative than raw accuracy.

4.5 Micro-enterprise exposure

Enterprises are assigned to exposure bands by distance to the nearest Metro Line 1 station (under 500 m, 500–1000 m, over 1000 m) and compared on the five Likert items by Kruskal–Wallis. An exploratory k-means clustering on nine standardised attributes identifies vulnerability profiles; k is chosen by silhouette and the silhouette values are reported so the reader can judge how weak the structure is. Consistent with the sample size, no multivariate model is fitted to the enterprise data.

4.6 Diagnostic-gap comparison and institutional benchmark

Officials' explanations for residents' non-use are coded into the same four analytic classes as the residents' own answers and compared by χ^2 . Because the enterprise sample is small, the enterprises' rating of SME support policy is benchmarked against Ho Chi Minh City's position on the Business

Support Policy sub-index of the Provincial Competitiveness Index for 2021 to 2024, and on the reorganised nine-index frame introduced in 2025.

5 Results

5.1 Modal split and barrier structure

Table 1 reports the modal split. A private motorcycle is the main mode for 384 of 460 respondents (83.5%). The bus is the main mode for eight (1.7%). Two-wheel electric vehicles account for 3.3%, a figure worth recording as a baseline for tracking electrification independently of mode shift.

Table 1. Main travel mode, resident survey (n = 460).

Main mode	n	%
Private motorcycle	384	83.5
Other / recoded	21	4.6
Private two-wheel EV	15	3.3
Private car	13	2.8
Bicycle	12	2.6
Bus	8	1.7
Ride-hail / taxi	5	1.1
Private electric car	2	0.4

Table 2 and Figure 1 report the barrier structure. The habit of using a motorcycle is chosen by 313 respondents, 68% of the full sample (95% CI 63.6–72.1) and 70.2% of non-users. The two infrastructural options together account for 51 respondents, 11.1% (95% CI 8.5–14.3). The intervals do not approach each other. The ratio of 6.1:1 is the paper’s headline quantity.

Table 2. Stated reason for not using public transport, with Wilson 95% confidence intervals.

Stated reason	Class	n	% of 460	95% CI	% of 446
Motorcycle habit	Behavioural	313	68.0	[63.6, 72.1]	70.2
Walking discomfort	Experiential	71	15.4	[12.4, 19.0]	15.9
Stop density / frequency	Infrastructure	29	6.3	[4.4, 8.9]	6.5
No parking at stops	Infrastructure	22	4.8	[3.2, 7.1]	4.9
Uses public transport	Not a barrier	14	3.0	[1.8, 5.0]	—
Bus quality / staff	Service	11	2.4	[1.3, 4.2]	2.5

The 446 denominator excludes the 14 respondents who selected "I do use public transport", an option erroneously included in the list of reasons for non-use.

The behavioural class is modal in every subgroup examined, but its share is not uniform (Table 3). Barrier class is independent of sex ($p = 0.431$), household income ($p = 0.069$) and commute time ($p = 0.142$), but not of education ($\chi^2(18) = 71.35$, $p < 0.001$, $V = 0.231$) or age tercile ($\chi^2(6) = 17.11$, $p = 0.009$, $V = 0.139$). The gradient runs in a specific direction: among respondents with lower-secondary or upper-secondary education the behavioural share is 81% and 84%, whereas among respondents with a college diploma or a university degree it is 44% and 51%, with the difference

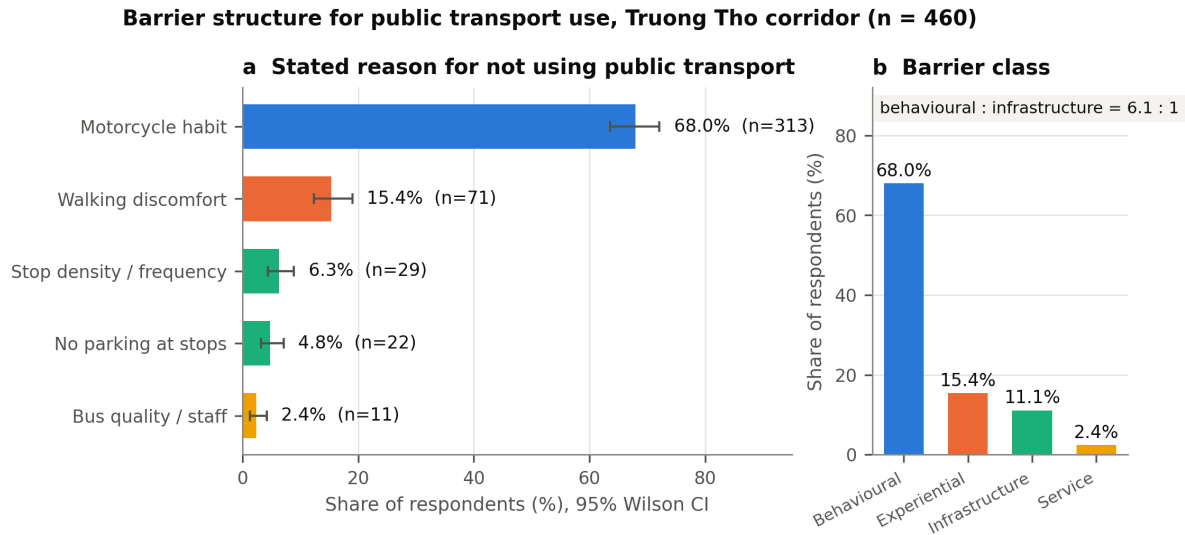


Figure 1. Barrier structure for public-transport use. (a) Stated reason with Wilson 95% intervals; (b) aggregation into four analytic classes. Behavioural attribution exceeds infrastructural attribution by about six to one.

absorbed mainly by the infrastructural and experiential classes. We return to this in Section 6.3, because the accessibility results in Section 5.4 show that the groups do not differ in the conditions they actually face — which makes the education gradient a gradient in articulation rather than in exposure.

Table 3. Independence of barrier class from respondent characteristics (χ^2 tests, non-users only).

Grouping	n	χ^2	df	p	Cramér's V	% behavioural (range)
Household income (5 bands)	446	19.92	12	0.069	0.122	48–80
Education (7 levels)	446	71.35	18	<0.001	0.231	43–84
Age tercile	446	17.11	6	0.009	0.139	59–76
Sex	446	2.75	3	0.431	0.079	67–73
Commute time	361	13.48	9	0.142	0.112	52–74

5.2 What the two snapshots can and cannot support

The comparability audit produced a result that reshaped the analysis, and it is reported before any before–after comparison because it constrains all of them. All 14 Metro Line 1 stations are present in the 1 January 2023 extract. They were mapped while the line was under construction. Matching the two snapshots station by station (Table 4), every station is present in both, and the largest positional displacement between snapshots is 31.8 m — cartographic refinement, not construction. Metro station geometry therefore carries no before–after signal whatsoever, and the natural-experiment design that a naive reading of the two extracts would suggest is not available.

Two stations were renamed between snapshots: Suoi Tien station → National University; East Bus Station → Suoi Tien Terminal.

Table 4. Metro Line 1 station audit across the two OpenStreetMap snapshots.

Station (2026 name)	In 2023 extract	In 2026 extract	Positional shift (m)
An Phu	yes	yes	22.7
Ba Son	yes	yes	6.9
Ben Thanh	yes	yes	6.4
Binh Thai	yes	yes	12.1
High Tech Park	yes	yes	13.3
National University	yes	yes	31.4
Opera House	yes	yes	4.7
Phuoc Long	yes	yes	31.8
Rach Chiec	yes	yes	17.7
Suoi Tien Terminal	yes	yes	16.4
Tan Cang	yes	yes	4.0
Thao Dien	yes	yes	15.8
Thu Duc	yes	yes	22.5
Van Thanh Park	yes	yes	17.5

What does change is the density of mapped bus and platform objects, which rises from 894 to 2324, an increase of 160% (Table 5, Figure 12). Inspecting the composition of that growth shows it is concentrated in objects tagged `public_transport=platform`, which rose from 677 to 2183, while objects tagged `highway=bus_stop` actually fell from 128 to 24. That signature is characteristic of a retagging and route-relation mapping campaign rather than of physical construction. Seventeen electric feeder routes were in fact opened to serve the metro, so genuine new infrastructure exists; but the two effects cannot be separated in these data, and any accessibility gain computed from the snapshots must be read as change in mapped accessibility, mixing real provision with rising completeness.

Table 5. OpenStreetMap completeness audit within the study window.

Object class	1 Jan 2023	1 Jan 2026	Change	Change %
Metro Line 1 stations	14	14	+0	+0.0
Bus/PT stop objects	894	2,324	+1,430	+160.0
of which <code>highway=bus_stop</code>	808	2,207	+1,399	+173.1
of which <code>public_transport=platform</code>	677	2,192	+1,515	+223.8
Commercial POIs	7,456	9,255	+1,799	+24.1
Highway ways	57,933	66,990	+9,057	+15.6

5.3 Measured accessibility at the two dates

Table 6 reports the accessibility variables. Median straight-line distance to the nearest mapped stop falls from 290 m to 140 m, and median network walking distance from 425 m to 238 m. Both changes are large and, on a paired test, unambiguous — but subject to the caveat above. The detour ratio moves in the opposite direction, from 1.42 to 1.60, which is the expected arithmetic consequence of denser stop mapping: the straight-line denominator falls faster than the network numerator when the newly mapped stop is on the far side of a block. Median distance to the nearest Metro Line 1 station is 1612 m and is unchanged between snapshots by construction; 112

Snapshot comparison and its completeness caveat

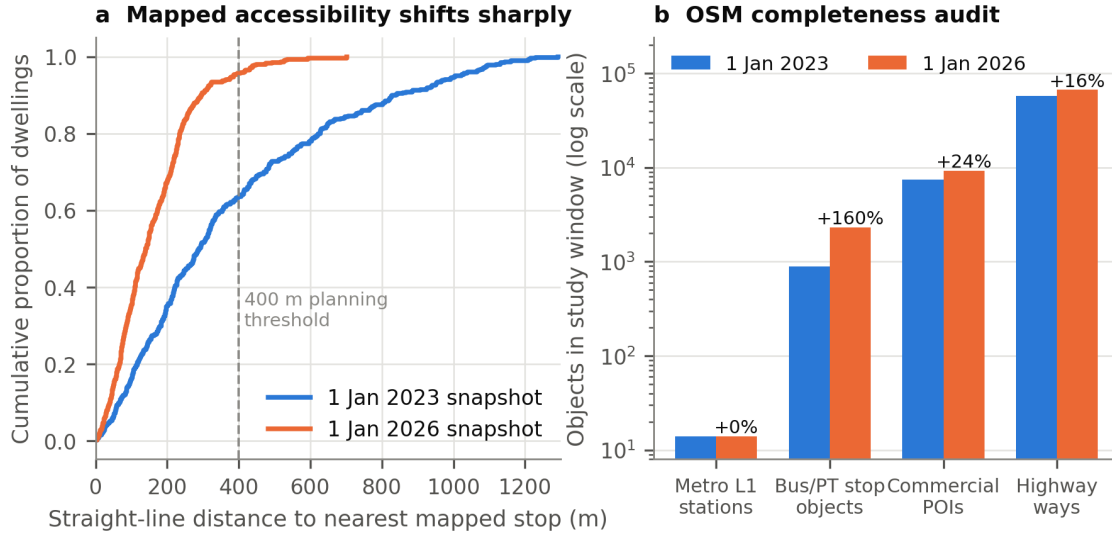


Figure 12. (a) Cumulative distribution of distance from dwelling to the nearest mapped stop under each snapshot. (b) Growth in each class of mapped object. Metro stations do not change; bus and platform objects grow by 160%.

respondents (24.4%) live within 1 km of a station.

Table 6. Accessibility variables at the two snapshot dates ($n = 459$ residents).

Variable	2023 median [IQR]	2026 median [IQR]	Median change	Wilcoxon p
Straight-line distance to nearest PT stop (m)	289.60 [145.42–547.70]	139.74 [75.34–224.65]	-98.02	1.53e-57
Network walking distance to nearest PT stop (m)	424.55 [227.98–866.68]	237.54 [125.82–370.95]	-96.08	4.97e-44
Detour ratio (network / straight-line)	1.42 [1.29–1.66]	1.60 [1.39–2.00]	0.14	1.85e-15
Commercial POIs within 1 km	73.00 [22.50–88.00]	76.00 [27.00–91.00]	3.0	9.37e-72
Straight-line distance to nearest Metro Line 1 station (m)	1605 [1006–1928]	1612 [1013–1933]	0.64	n.a. (stations mapped in both snapshots)

5.4 Central test A: stated barriers against measured conditions

This is the paper’s decisive test, and it uses a single snapshot, so the completeness problem does not touch it. If the barriers residents name describe physical conditions, then those who name infrastructure should live further from stops, walk further, or face more circuitous routes than those who name habit. They do not (Table 7, Figure 3).

Across the five stated-reason groups, 2023 network walking distance does not differ significantly ($H = 6.70$, $p = 0.153$, $\varepsilon^2 = 0.006$), nor does the 2023 detour ratio ($H = 5.70$, $p = 0.223$, $\varepsilon^2 = 0.004$), nor does 2023 straight-line distance ($H = 8.48$, $p = 0.075$, $\varepsilon^2 = 0.010$). The point estimates are ordered plausibly — those citing stop density have the longest median walk, 551 m against 388 m for the habit group — but the distributions overlap almost completely and the effect sizes are negligible by any convention. Neither commercial density within 1 km ($p = 0.135$) nor road

Mapped public-transport accessibility around the Truong Tho corridor, 2023 vs 2026

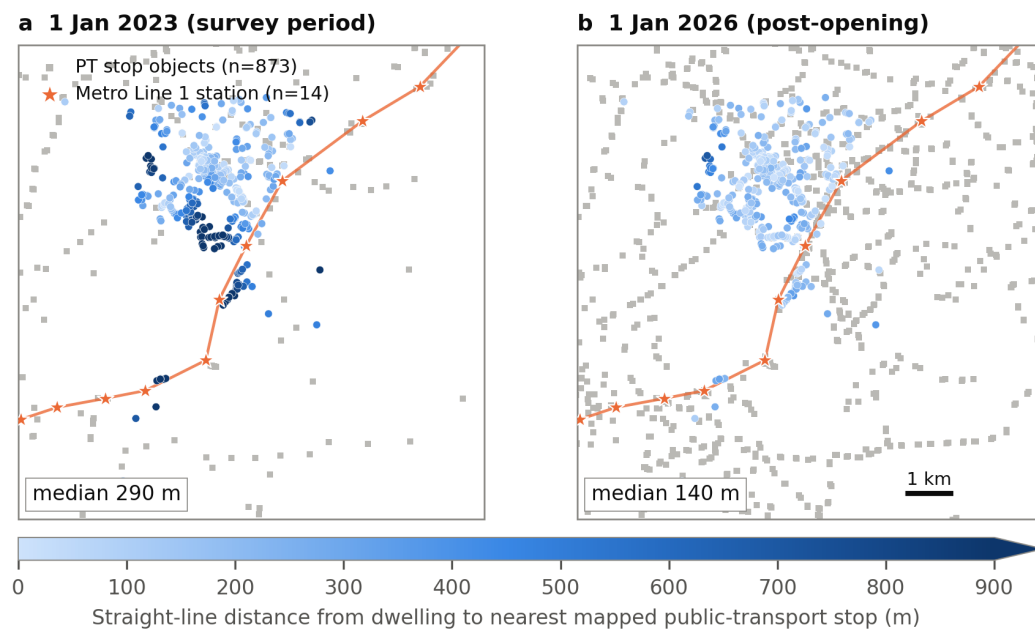


Figure 2. Mapped public-transport accessibility around the Truong Tho corridor. Each point is a surveyed dwelling, coloured by distance to the nearest mapped stop; grey squares are mapped stop objects; stars are the fourteen Metro Line 1 stations, present in both snapshots.

density within 500 m ($p = 0.368$) differs. Only distance to the nearest Metro Line 1 station differs ($H = 19.02$, $p = 0.001$, $\varepsilon^2 = 0.034$), and inspection shows why: the small group citing the absence of parking at stops has a median distance of 882 m against 1676 m for the habit group, i.e. it is the group that lives closest to a future station and can therefore most concretely imagine the park-and-ride problem.

Table 7. Kruskal–Wallis tests of accessibility variables across stated-reason groups ($n = 445$).

Variable	H	df	p	ε^2	Habit	Walking	Stops	Parking	Quality
2023 straight-line distance to nearest PT stop (m)	8.48	4	0.075	0.010	270.03	293.36	413.36	289.60	198.53
2023 network walking distance to stop (m)	6.70	4	0.153	0.006	388.01	450.13	550.78	415.14	436.07
2023 detour ratio	5.70	4	0.223	0.004	1.42	1.43	1.43	1.33	1.58
Distance to nearest Metro Line 1 station (m)	19.02	4	0.001	0.034	1676.31	1437.08	1453.11	881.62	1447.86
Change in mapped PT accessibility 2023-2026 (m)	11.61	4	0.021	0.017	89.71	95.16	234.45	146.15	27.23
Commercial POIs within 1 km (2023)	7.02	4	0.135	0.007	74.00	76.00	38.00	79.00	92.00
Road density within 500 m (m/km ²)	4.29	4	0.368	0.001	22165.70	22470.97	23710.91	21512.49	21674.84

Group sizes: motorcycle habit $n = 313$, walking discomfort $n = 71$, stop density/frequency $n = 29$, no parking at stops $n = 21$, bus quality $n = 11$. Cell entries are group medians.

Objective accessibility does not differ by the barrier respondents name ($n = 445$; Kruskal–Wallis)

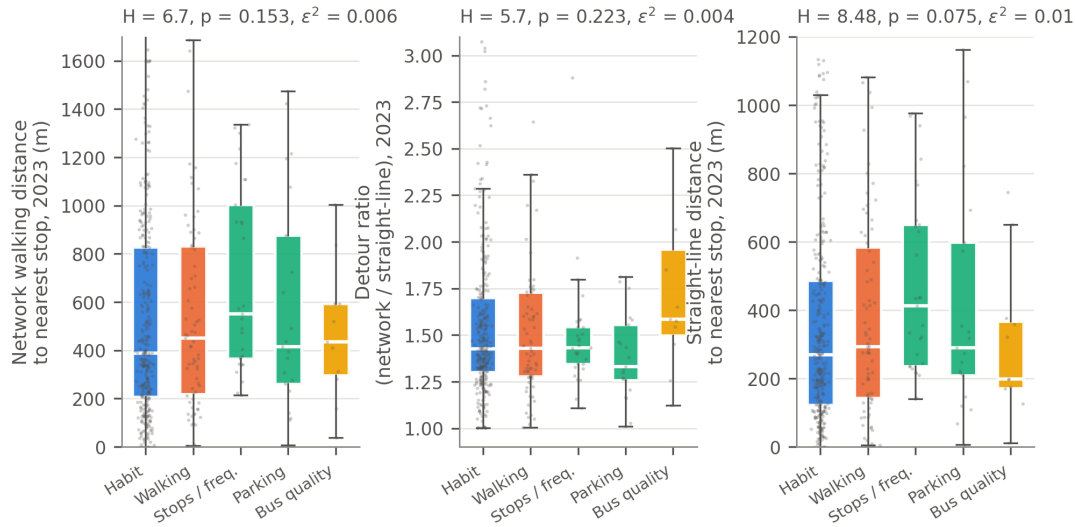


Figure 3. Objective accessibility by the barrier respondents name. Three independent measures, one snapshot, no significant difference in any of them. This is the paper's central result.

5.5 Central test C: the walking-discomfort group

The 71 respondents (15.4%) who say they are uncomfortable walking to stops occupy the analytically pivotal position, because their answer sits between a physical condition and a preference. If

their walks are objectively worse, the barrier is a pavement problem with a concrete and affordable remedy. If not, the answer belongs with habit and the headline ratio changes.

Table 8 and Figure 4 settle it. Against all other barrier groups combined, this group's 2023 network walking distance (median 450 m vs 398 m, $U = 13,449$, $p = 0.835$), its 2023 detour ratio (1.43 vs 1.43, $p = 0.977$), its straight-line distance ($p = 0.732$) and its local road density ($p = 0.235$) are all statistically indistinguishable. Rank-biserial effect sizes are 0.03 or smaller. The only measure approaching conventional significance is the 2026 detour ratio (1.73 vs 1.58, $p = 0.082$), which is a property of the later snapshot's stop mapping rather than of the respondents.

We therefore reclassify this group as perceptual. Doing so raises the behavioural-plus-perceptual share to 83.4% of the sample against an unchanged infrastructural 11.1%, a ratio of about 7.5:1. This is the sense in which the paper's title is meant literally: what residents describe as a walking problem is not, on measurement, a walking problem.

Table 8. Walking-discomfort group against all other barrier groups (Mann–Whitney).

Variable	Walking group (n = 71)	All other groups	U	p	Rank-biserial r
2023 straight-line distance to stop (m)	293.36	276.28	13,618	0.732	0.026
2023 network walking distance (m)	450.13	398.05	13,449	0.835	0.016
2023 detour ratio	1.43	1.43	13,212	0.977	-0.002
2026 network walking distance (m)	251.81	236.39	13,303	0.98	0.002
2026 detour ratio	1.73	1.58	15,003	0.082	0.130
Road density within 500 m (m/km ²)	22470.97	22089.61	14,458	0.235	0.089

Residents who say walking to stops is uncomfortable do not in fact walk further

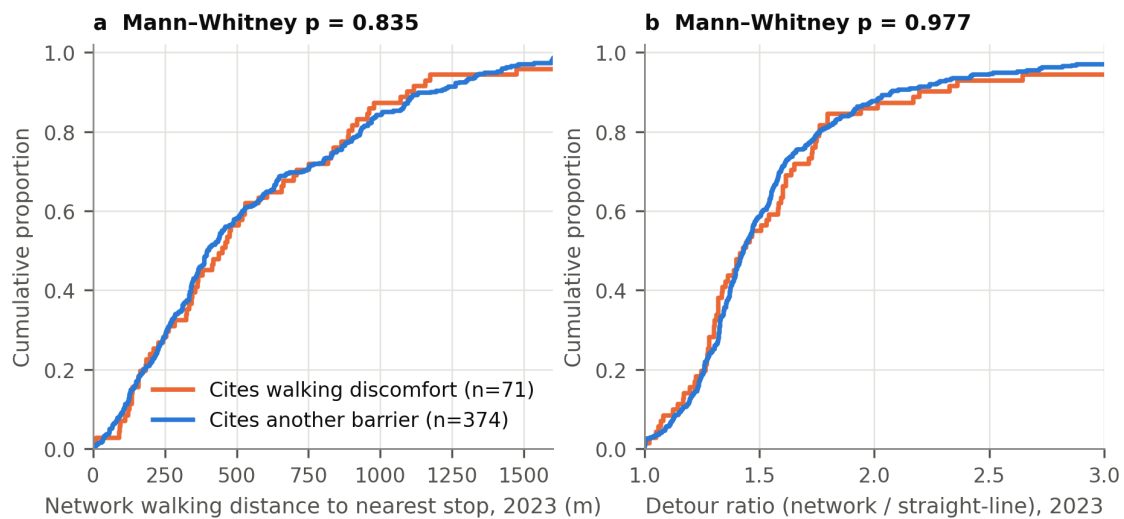


Figure 4. Cumulative distributions of network walking distance and detour ratio for residents who cite walking discomfort against all other barrier groups. The curves are superimposed.

5.6 Central test B: who received the accessibility gain

The change in mapped accessibility does differ across groups ($H = 11.61$, $p = 0.021$, $\varepsilon^2 = 0.017$), and in the direction that would be expected of targeted investment: the group citing stop density and frequency gained a median of 234 m against 90 m for the habit group and 27 m for those citing bus quality (Figure 5). Read literally, provision and mapping have concentrated on the population that named the deficit.

Two qualifications matter. First, the audit in Section 5.2 means this gain mixes real provision with rising completeness, and there is no way to separate them at respondent level. Second, and more consequentially, the group that gained most is 11.1% of the sample. Even on the most generous reading, in which every metre of the gain is genuine new infrastructure precisely targeted at the people who asked for it, the intervention addresses about one ninth of the stated barrier structure while the dominant two thirds is untouched. Targeting efficiency and policy sufficiency are different questions, and this corridor answers them differently.

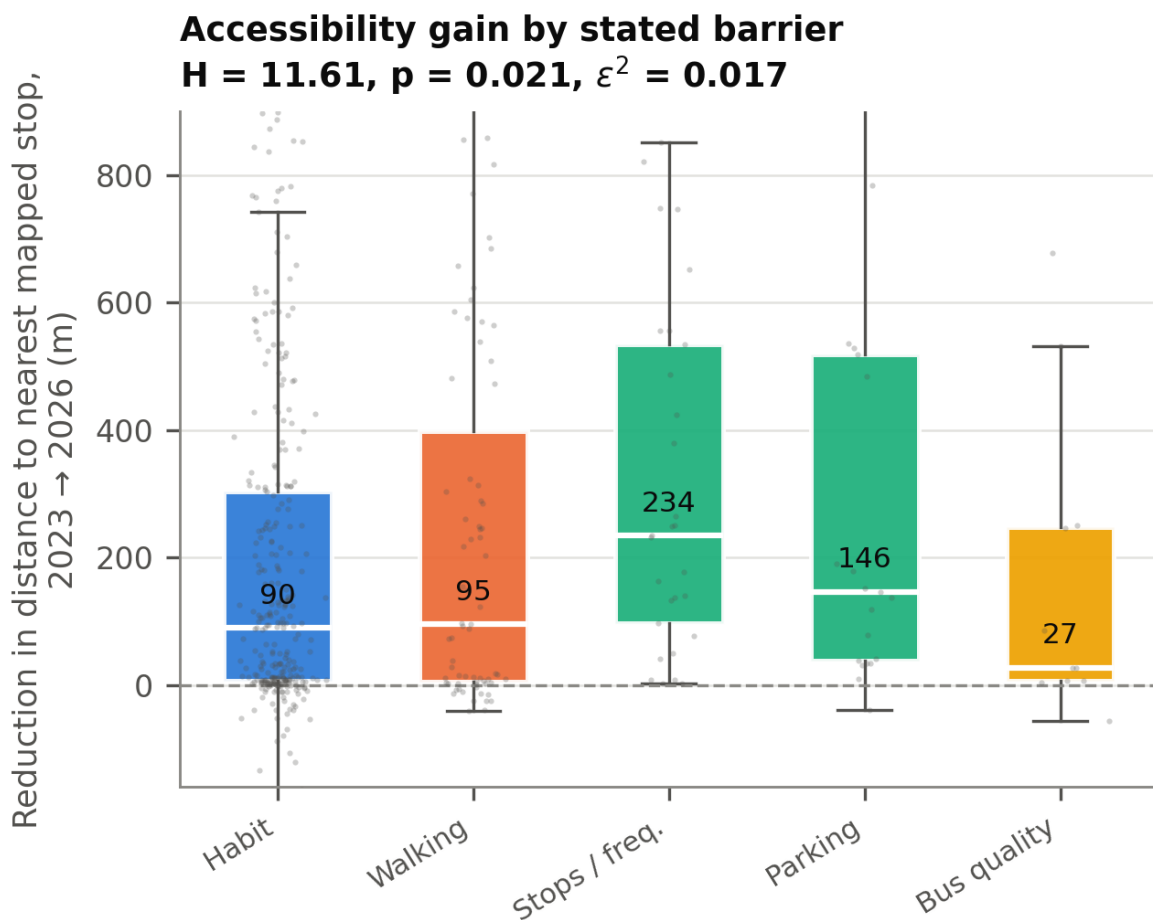


Figure 5. Reduction in distance to the nearest mapped stop between snapshots, by stated barrier. The infrastructure group gains most — but it is 11.1% of the sample.

5.7 Predicting motorcycle dependence

The logistic model ($n = 368$ complete cases, McFadden pseudo- $R^2 = 0.094$, likelihood-ratio $p = 0.0042$) returns two significant coefficients, and neither is an accessibility variable. Household income is positive and significant: each step up the five-band income scale multiplies the odds of motorcycle dependence by 1.76 (95% CI 1.19–2.62, $p = 0.005$). Age is weakly negative (OR 0.975 per year, $p = 0.041$); younger residents are more motorcycle-dependent. Every spatial predictor is null — distance to the nearest stop ($p = 0.538$), network walking distance ($p = 0.634$), detour ratio ($p = 0.562$), distance to the nearest metro station ($p = 0.389$), commercial density ($p = 0.720$) and road density ($p = 0.224$).

The income result deserves emphasis because it is the opposite of the pattern a deprivation account would predict, and it was not the result we expected. Motorcycle dependence here is not the constrained fallback of households that cannot afford anything else; it rises with the capacity to pay. That removes the most common alternative explanation for the habit finding and strengthens the lock-in reading: as incomes rise within this regime, households buy more deeply into it rather than out of it.

Table 9. Logistic regression of motorcycle dependence ($n = 368$).

Predictor	β	SE	OR	95% CI	p
Household income (5 ordered bands)	+0.567	0.201	1.76	[1.19, 2.62]	0.005
Education (7 ordered levels)	-0.038	0.125	0.96	[0.75, 1.23]	0.762
Age (years)	-0.025	0.012	0.97	[0.95, 1.00]	0.041
Female (ref: male)	-0.195	0.336	0.82	[0.43, 1.59]	0.563
Own-account business owner (ref: other)	+0.484	0.377	1.62	[0.78, 3.39]	0.199
Commute time (4 ordered bands)	+0.021	0.191	1.02	[0.70, 1.48]	0.914
Distance to nearest PT stop, 2023 (per 100 m)	-0.173	0.282	0.84	[0.48, 1.46]	0.538
Network walking distance to stop, 2023 (per 100 m)	+0.092	0.194	1.10	[0.75, 1.60]	0.634
Detour ratio, 2023 (per 0.1)	+0.027	0.047	1.03	[0.94, 1.13]	0.562
Distance to nearest Metro Line 1 station (per 100 m)	+0.023	0.027	1.02	[0.97, 1.08]	0.389
Commercial POIs within 1 km (per 10)	-0.017	0.047	0.98	[0.90, 1.08]	0.72
Road density within 500 m (per 1000 m/km ²)	-0.075	0.062	0.93	[0.82, 1.05]	0.224

The cross-validated model comparison (Table 10) shows that this near-nullity is not an artefact of the linear specification. Under identical stratified 5-fold cross-validation, logistic regression reaches ROC-AUC 0.635, random forest 0.623 and gradient boosting 0.61; no model exceeds the majority-class accuracy of 0.853. Permutation importance on the best model attributes essentially all of the modest discrimination to income (mean AUC drop 0.100) and age (0.042), with every accessibility variable below 0.020 and within one standard deviation of zero (Figure 7b).

We read the weak discrimination as substantively informative rather than dismissing it as a mod-

Table 10. Cross-validated comparison of three classifiers for motorcycle dependence.

Model	ROC-AUC	AUC SD (folds)	Accuracy	Majority class	Brier
Logistic regression	0.635	0.038	0.853	0.853	0.123
Random forest	0.623	0.035	0.812	0.853	0.146
Gradient boosting	0.610	0.073	0.832	0.853	0.135

elling failure. A behaviour adopted by 83.5% of a population, and only weakly predicted by income, education, sex, commute length or the measured built environment, operates as the corridor’s dominant baseline rather than as a sharply segmented choice. Consistent with recent comparisons of machine-learning and choice models, however, predictive performance alone cannot establish the behavioural mechanism or exclude omitted heterogeneity (Hillel et al., 2021; Martín-Baos et al., 2023; Dahmen et al., 2024).

5.8 Micro-enterprise exposure

The enterprise sample is uniformly micro and uniformly consumer-facing (Table 11). 75 of 80 employ five people or fewer; 70 of 80 sell exclusively to consumers; groceries and food-and-drink service together account for 50 of 80. Fifty-seven per cent serve fewer than 100 customers a month, and 30 of 80 have traded for under two years, indicating high turnover in a thin-margin population. Median distance to the nearest Metro Line 1 station is 1239 m; twelve enterprises are within 500 m and eighteen between 500 and 1000 m (Figure 9a).

Table 11. Micro-enterprise profile (n = 80).

Characteristic	n	% of 80
Micro (0–5 staff)	75	93.8
Grocery	26	32.5
Food & drink service	24	30.0
B2C only	70	87.5
<100 customers/month	46	57.5
Trading <2 years	30	37.5
Self-rated growth ‘fair’ or better	63	78.8

The perception battery reproduces a paradox worth stating plainly. Enterprises do not believe they hold a competitive advantage where they are ($M = 3.04$, $SD = 0.77$, i.e. neutral), yet they intend to stay ($M = 3.69$) and believe the area is changing for the better ($M = 3.71$). Four of the five items are flat across exposure bands. The exception is the rating of SME support policy, which is higher near the alignment than away from it (3.58 and 3.61 against 3.10; $H = 6.60$, $p = 0.037$). With twelve enterprises in the nearest band this is a single significant result among five tests and should be treated as a hypothesis for follow-up, not a finding: it is consistent with more visible state activity near the works, and equally consistent with chance.

Exploratory clustering identifies four profiles (silhouette 0.22, i.e. weak separation), of which two are substantively interesting: a group of fifteen high-footfall enterprises close to the alignment in the densest commercial fabric, and a group of ten younger, lower-volume, non-exclusively-B2C enterprises with the lowest intention to remain. The clustering is reported for completeness and we

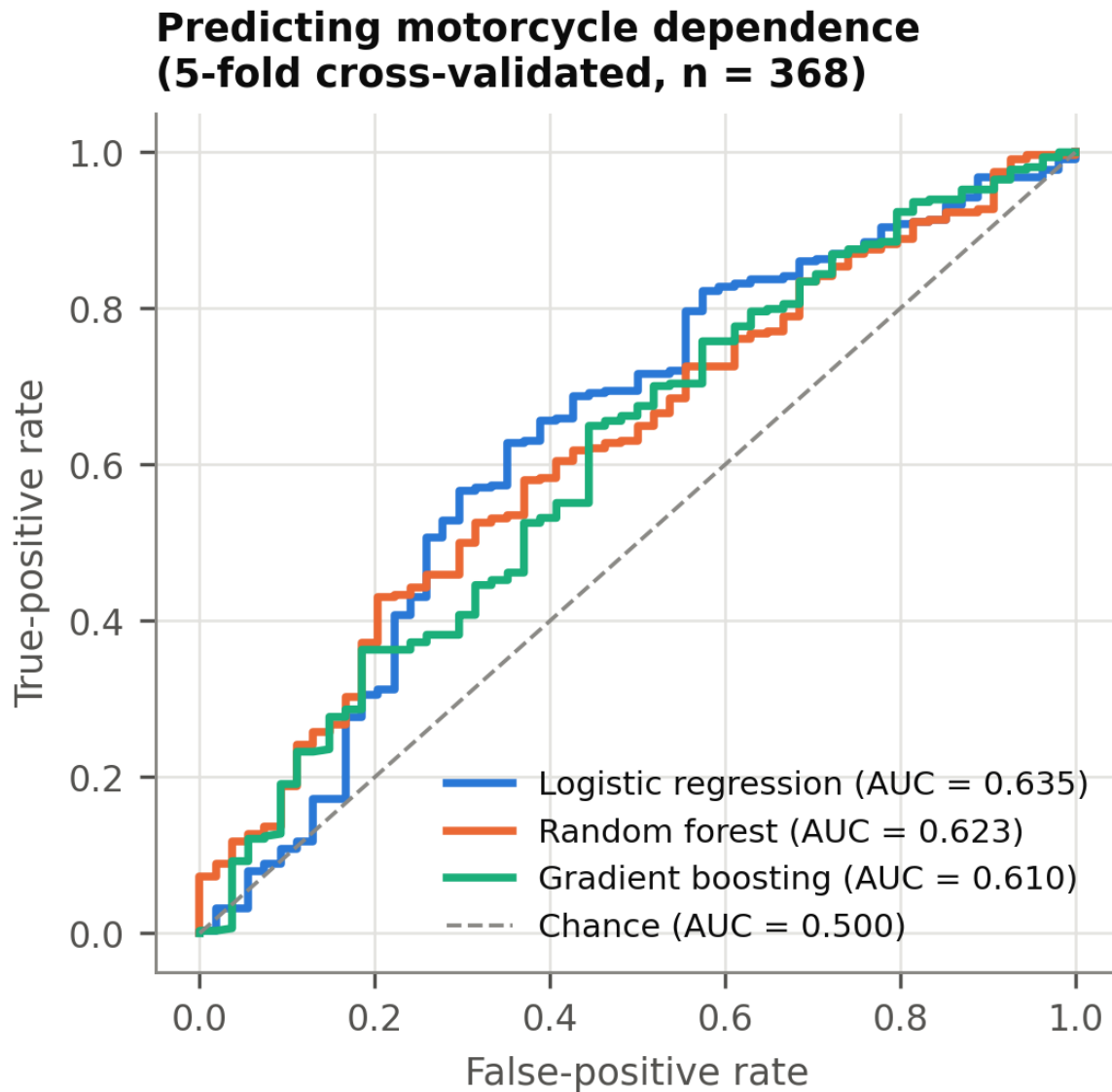


Figure 6. Cross-validated ROC curves. All three models sit close to the diagonal.

Table 12. Micro-enterprise perceptions by distance to the nearest Metro Line 1 station.

Item	M	SD	<500 m (12)	500–1000 m (18)	>1000 m (50)	H	p
Q4.7 My business has a competitive advantage here	3.04	0.77	3.08	2.94	3.06	0.72	0.698
Q4.8 I intend to keep trading at this location	3.69	0.67	3.67	3.83	3.64	1.21	0.545
Q4.9 Changes in this area are positive	3.71	0.56	3.67	3.67	3.74	0.39	0.824
Q4.10 SME support policy is effective	3.29	0.87	3.58	3.61	3.10	6.60	0.037
Q4.11 Legal/administrative support is adequate	3.05	0.69	3.08	3.06	3.04	0.09	0.956

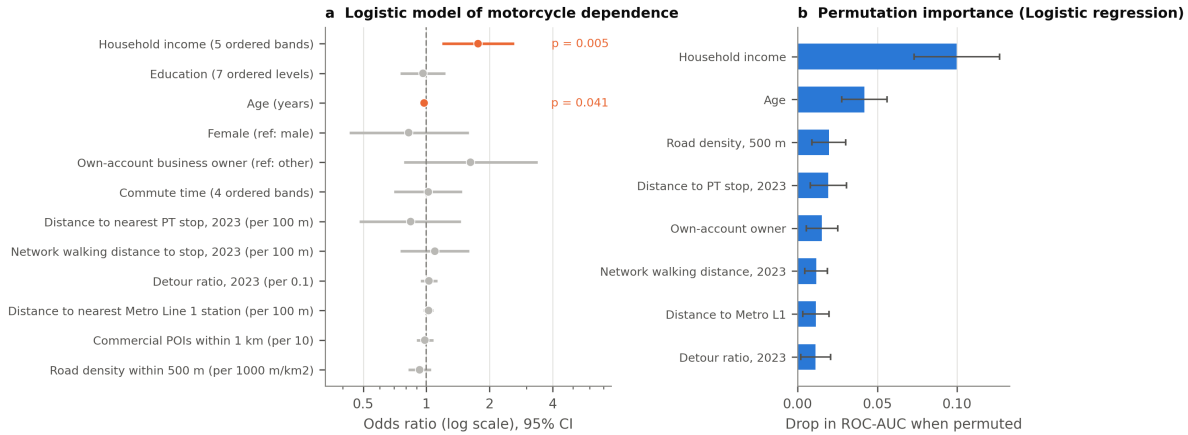


Figure 7. (a) Odds ratios with 95% intervals from the logistic model; significant coefficients in orange. (b) Permutation importance for the best-performing model.

draw no inference from it.

5.9 The diagnostic gap

Residents and officials were asked the same question about the same population. The answers diverge almost completely (Table 13, Figure 8). Residents attribute non-use to habit 70.2% of the time and to service or system attributes 2.5% of the time. Officials invert the ratio: 12 of 21 who expressed a view (57.1%) attribute non-use to journeys taking too long, while 4 (19.0%) point to a weak bus-riding culture. The distributions differ sharply ($\chi^2(3) = 129.56$, $p < 0.001$), with a gap of 51.1 percentage points on the behavioural class and 54.7 points in the opposite direction on the service class.

Table 13. Attribution of non-use: residents against local officials.

Barrier class	Residents n	Residents %	Officials n	Officials %	Gap (pp)
Behavioural	313	70.2	4	19.0	+51.1
Experiential	71	15.9	3	14.3	+1.6
Infrastructure	51	11.4	2	9.5	+1.9
Service	11	2.5	12	57.1	-54.7

Residents n = 446 (non-users); officials n = 21 of 23, excluding two who expressed no view.

This is not a claim that officials are careless. Journey time is a real and rational concern, and it is the variable an official can act on with the instruments an official holds. But the population being governed says something else, and the instrument implied by each diagnosis is different: journey time calls for more service, higher frequency and priority lanes; habit calls for interruption of a routine — trial periods, targeted personalised planning, employer schemes, workplace and school travel plans. A diagnosis of the second kind is unlikely to emerge from a process designed around the first.

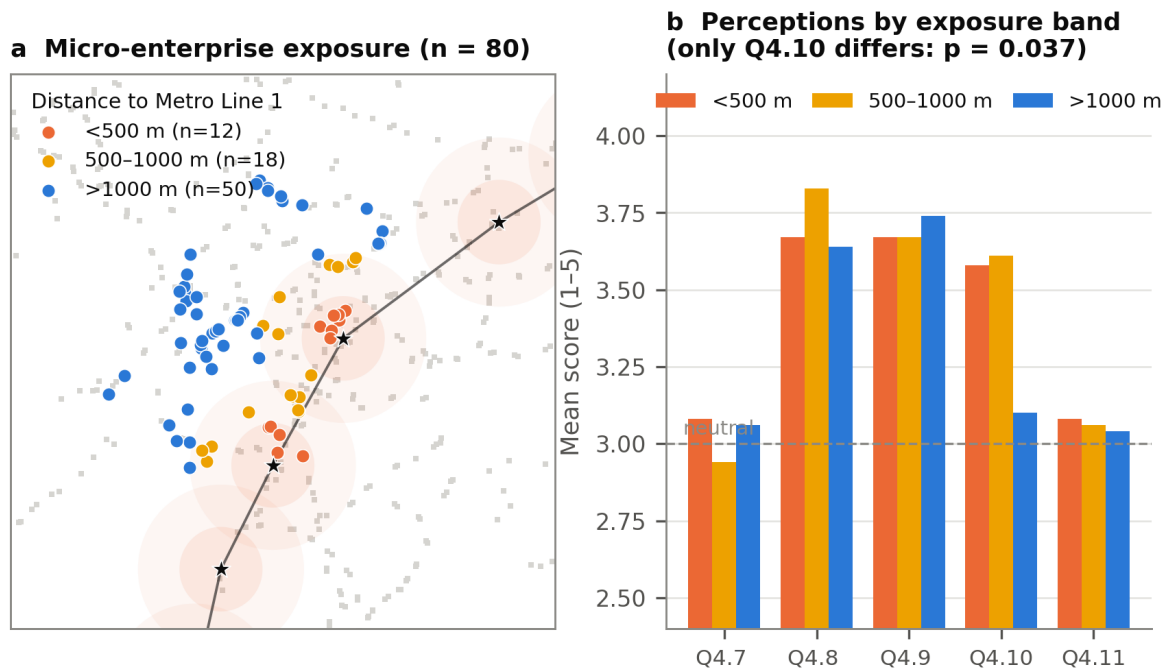


Figure 9. (a) Micro-enterprise locations by exposure band, with 500 m and 1000 m station buffers. (b) Perception items by exposure band; only the SME support-policy item differs.

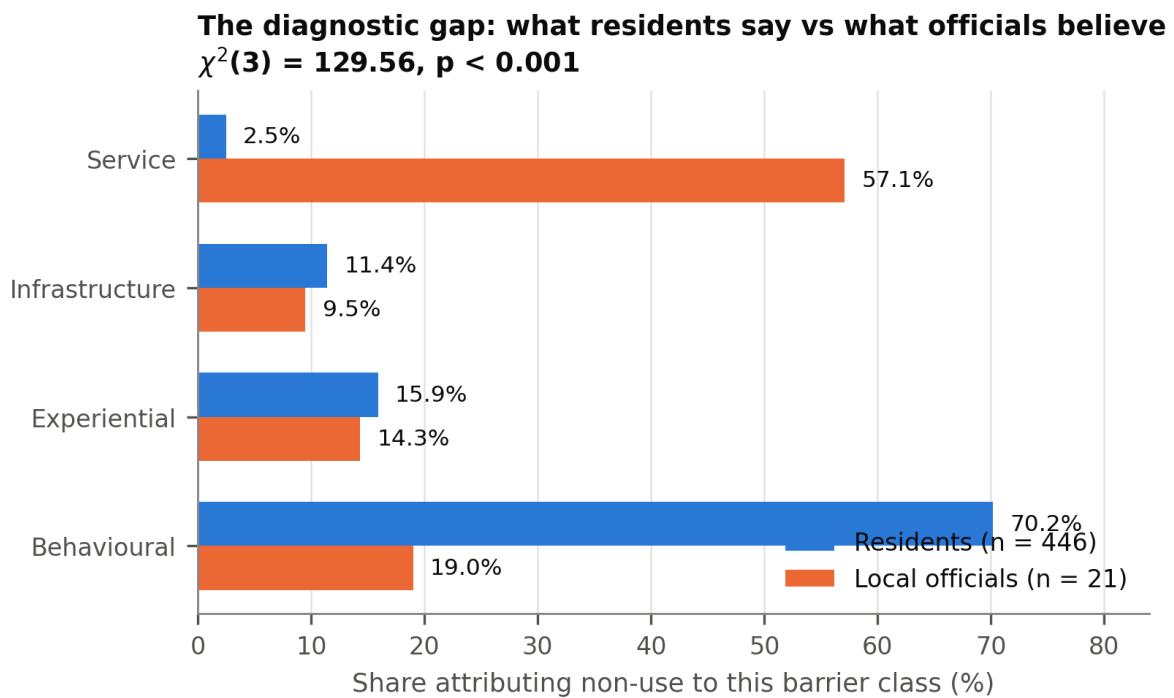


Figure 8. Attribution of non-use by residents and by local officials, in the same four analytic classes.

5.10 Institutional benchmark

The enterprises' neutral rating of SME support policy ($M = 3.29$, $SD = 0.87$) is consistent with the province's own competitiveness record rather than idiosyncratic to this sample. Ho Chi Minh City's score on the Business Support Policy sub-index of the Provincial Competitiveness Index has stayed above the national median throughout 2021 to 2024, but its rank has fallen from 2nd of 68 provinces in 2021 to 13th of 68 in 2024, and it stands 14th of 37 provinces scored under the reorganised nine-index frame in 2025 (Table 14, Figure 11). A locally neutral rating in a province that is slipping relative to its peers is the expected reading, and it makes the small enterprise sample an illustration of a general pattern rather than an outlier.

Table 14. Ho Chi Minh City on the PCI business-support sub-index, 2021–2025 (VCCI).

Year	Sub-index	HCMC	National median	Rank	Provinces scored
2021	Business Support Policy	8.54	6.85	2	68
2022	Business Support Policy	7.04	5.82	2	68
2023	Business Support Policy	7.37	6.45	8	67
2024	Business Support Policy	7.81	7.18	13	68
2025	Business & labour support (new 9-index frame)	5.61	5.43	14	37

Institutional benchmark: provincial business-support performance (PCI, VCCI)

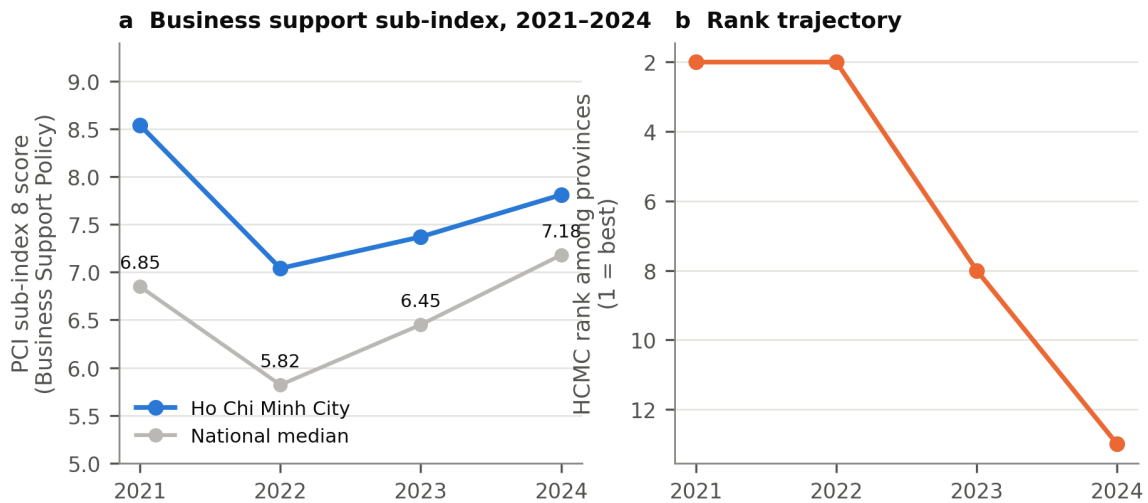


Figure 11. (a) Business-support sub-index score for Ho Chi Minh City against the national median. (b) Rank trajectory.

5.11 Built-environment change along the corridor

Three remote-sensing sources were read in rings around the fourteen stations (Table 15, Figure 10), and the result is a caution rather than a finding. GHS-POP shows growth of 13.2% to 13.7% between the 2020 and 2025 epochs in every ring, with no gradient. This uniformity is a property of the product, not of the corridor: the 2025 epoch is a modelled projection, not an observation, and it cannot evidence differential transit-oriented growth. GHS built-up surface does show a gradient, but the inverse of the transit-oriented development expectation: growth of 0.13% within 500 m of a station against 1.40% at 1500–2000 m. The most parsimonious reading is saturation — the

innermost rings are the corridor’s oldest and densest fabric and have little room left, whereas the outer rings are still filling in. The Sentinel-2 built-up index change is slightly higher near the stations (median 0.0064 against 0.0045 at 1500–2000 m), which is consistent with redevelopment intensity rather than footprint expansion. None of these differences is large, and we report them to establish that no dramatic corridor transformation had occurred by the 2026 snapshot, not to claim a transit-oriented development effect.

Table 15. Built-environment change by distance ring from Metro Line 1.

Ring (m)	GHS-POP change (%)	GHS BUILT-S change (%)	Median Δ NDBI	2020 population in ring
0–500	13.23	0.132	0.0064	216,804
500–1000	13.31	0.526	0.0056	374,899
1000–1500	13.55	0.792	0.0054	416,710
1500–2000	13.73	1.402	0.0045	405,730
2000–3000	13.53	1.354	0.0057	841,691

Built-environment change by distance ring from Metro Line 1

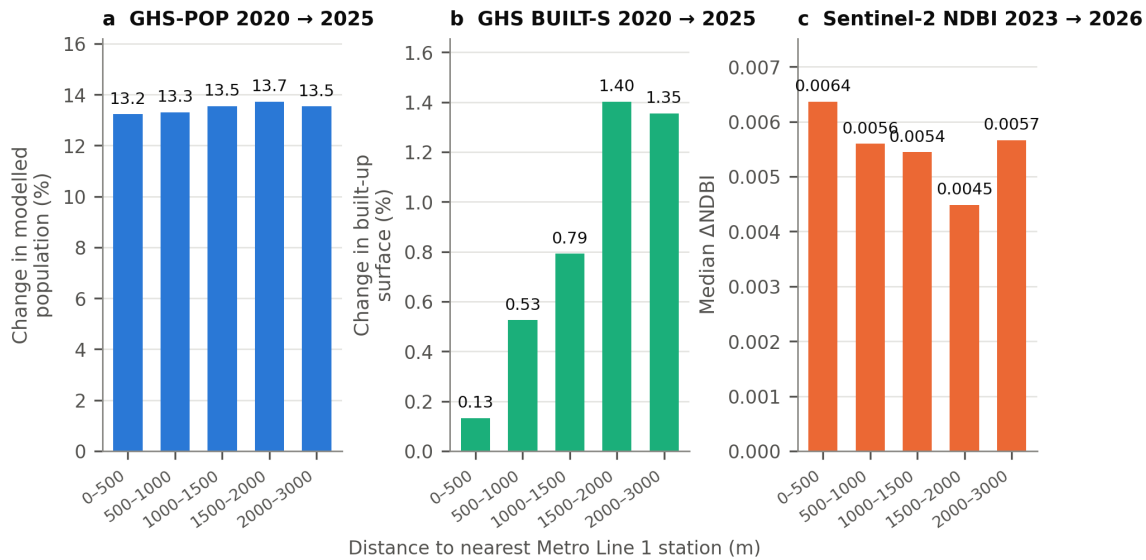


Figure 10. Built-environment change by distance ring from Metro Line 1, from three independent sources.

6 Discussion

6.1 Why an infrastructure-first strategy may under-deliver

The results describe a corridor in which the constraint on public-transport use is not the supply of public transport. Two thirds of residents name a routine; the remainder who name a physical condition turn out, on measurement, to face the same physical conditions as everybody else; no accessibility variable predicts motorcycle dependence; and three classifiers cannot distinguish motorcycle users from non-users better than the base rate. The habit literature anticipates this configuration precisely. Verplanken and Aarts (1999) and Gardner (2009) predict that in a stable context, mode choice ceases to be re-evaluated and becomes cue-driven; Fujii and Kitamura (2003)

show that even a substantial change in the alternative’s cost has effect only while it interrupts the routine.

The practical consequence is a transmission problem. Metro Line 1 changes the option set. It does not, by itself, create the occasion on which the option set is reconsidered. For a household that has not evaluated its mode choice in a decade, a new rail line 1.6 km away is information that never enters a decision, because no decision is being taken. This is why the 6.1:1 ratio matters more than the modal split: it locates the constraint at a point that capital expenditure does not reach.

The income result is what turns this from a psychological observation into a systemic one. If motorcycle dependence fell with income, we would expect it to erode as the corridor’s households grow richer, and the policy question would be one of patience. It rises with income (OR 1.76 per band). Within this regime, additional resources are spent on deeper participation in it — a better motorcycle, a second one, a more convenient parking arrangement — not on exit. That is the signature of lock-in in Unruh’s (2000) sense: a self-reinforcing configuration in which the returns to staying increase with the capacity to leave.

The low pseudo- R^2 and cross-validated AUC values nevertheless require a more cautious interpretation than universality alone. Weak discrimination can arise because an outcome is nearly ubiquitous, but it can also arise from omitted variables that the questionnaire did not measure. In particular, the resident instrument contains commute duration but no activity diary, trip-purpose sequence, escorting responsibility, household vehicle allocation, or need to carry passengers and goods. Hensher and Reyes (2000) identify trip chaining as a barrier to public-transport use because a tour that combines work, shopping, school escorting and other stops is evaluated as a whole rather than as a single home–work trip; Primerano et al. (2008) similarly show that chaining emerges from the interaction of household roles, activity schedules and transport constraints. Variables of this kind could distinguish motorcycle users from non-users even when residential accessibility does not. The models therefore show that the observed socio-demographic and spatial variables have little discriminatory power; they do not establish that no unobserved heterogeneity exists.

This qualification also changes how “habit” should be understood. In this corridor it may be psychological automaticity, but it may simultaneously be a learned adaptation to the multi-purpose organisation of household life. A motorcycle permits an adult to commute, divert through a market, carry purchases and collect a child within one flexible tour, often with another household member or cargo on the vehicle. A metro line can perform the trunk segment efficiently while still imposing access, waiting and transfer costs at every break in that chain. Repetition can then make a functionally useful arrangement automatic: the routine persists not only because travellers fail to reconsider it, but because it bundles tasks that the available public-transport journey does not yet integrate. Future models should therefore add travel diaries or GPS-based tour reconstruction, escorting and care responsibilities, passenger/cargo loads and household vehicle availability. Doing so would test whether the residual “habit” effect remains after the practical value of trip chaining is observed.

6.2 The motorcycle as economic infrastructure

The enterprise data supply the mechanism that makes the lock-in economic rather than merely cultural. Ninety-four per cent of the corridor’s surveyed enterprises employ five people or fewer, 88% sell exclusively to consumers, and the two largest sectors — groceries at 33% and food-and-drink service at 30% — are the archetypal drop-in trades. Their business geometry is a shopfront on a street where a customer can stop a motorcycle at the door, buy, and leave without parking. Kim (2012, 2015) describes this sidewalk economy in Ho Chi Minh City as a productive institution with its own property norms; Turner and Schoenberger (2012) document how quickly such livelihoods deteriorate when access is administratively reorganised.

This reframes what a mode shift would do. The transit-oriented development literature generally treats increased pedestrian footfall as a benefit to local retail, and for retail formats designed around walking it is. But a shift from riders who stop at the door to passengers who walk from a station is not a neutral substitution for a grocery whose entire layout, stock depth and staffing assume the former. The corridor’s enterprises appear to sense something of this without articulating it: they rate their own competitive advantage at exactly neutral ($M = 3.04$) while intending to remain ($M = 3.69$) and expecting the area to improve ($M = 3.71$). We cannot distinguish between speculative confidence in future location value and simple absence of alternatives; a qualitative follow-up is the natural way to separate them, and we flag it as the highest-value next step.

The theoretical claim we draw is that socio-technical lock-in should be specified at the level of the economic ecosystem, not the vehicle. The motorcycle in Ho Chi Minh City is not one element of a transport regime that could be swapped for another; it is the substrate on which a large part of the street-level economy is built. Geels (2012) treats regime change as multi-dimensional by construction, but empirical transport applications often operationalise the regime as vehicles, fuels and infrastructure. This case suggests that where the incumbent technology also organises retail form and livelihoods, the distributional dimension is not an add-on to the transition analysis but a constitutive part of it.

This is also where the Global South TOD literature changes the appraisal frame. Station-area accessibility and pedestrian connectivity are necessary conditions (Vale, 2015), but recent guidance for emerging cities treats institutional coordination, affordability, inclusion and protection from displacement as equally integral to TOD (World Bank, 2024). In a motorcycle-oriented commercial corridor, the relevant outcome is therefore not simply whether pedestrian footfall rises, but which businesses can convert that footfall into custom, which lose loading or stopping access, and which bear the transition costs.

6.3 Diagnosis before instrument choice

The 51-point gap between residents’ and officials’ attributions is, in policy terms, the most actionable result in the paper. Officials name travel time; residents name habit. Each diagnosis implies a different instrument, and only the first is reachable with the instruments a transport authority conventionally holds. The risk is not that the wrong thing gets built — more frequent, faster service is worth having on its own terms — but that the portfolio contains no instrument aimed at the barrier two thirds of the population reports.

The officials' travel-time diagnosis may also encode more than time spent moving. Public-transport users experience generalised journey time, including uncertain waiting, access and interchange, and these components are commonly valued more negatively than in-vehicle time. Wardman's (2004) review finds substantial penalties for waiting and interchange while cautioning against applying one universal multiplier, and Iseki and Taylor (2009) show that the burden of a transfer depends on reliability, information, safety, comfort and the transfer environment rather than on elapsed minutes alone. In the Line 1 corridor, an official complaint about "long journeys" could therefore partly be a complaint about irregular feeder headways, missed connections and the uncertainty attached to the first and last legs. Because the present data contain neither scheduled nor observed headways, this interpretation cannot be tested here. It does, however, imply that evaluating the feeder network requires observed waiting-time and reliability data, not only route coverage or nominal end-to-end travel time.

The education gradient in Section 5.1 supports this reading rather than complicating it. Respondents with tertiary education name infrastructural barriers roughly twice as often as those with secondary education, yet the accessibility tests show the two groups face indistinguishable conditions. The gradient is therefore in the vocabulary available for describing a barrier, not in the barrier itself. Consultation processes that recruit articulate participants will over-sample infrastructural framings for exactly this reason, which is a plausible route by which an official diagnosis drifts away from the population's own.

6.4 What a combined package would look like

We take the policy implication to be additive rather than substitutive. The infrastructure is built and its feeder network is being extended; that work should continue. What the baseline argues for is a parallel behavioural component sized to the two thirds of the barrier structure that infrastructure does not address. The habit literature is unusually specific about what works: interruption at moments when routines are already broken (Verplanken et al., 1998; Schwanen et al., 2012), time-limited free or discounted trial periods (Fujii and Kitamura, 2003), personalised travel planning, and workplace and school travel plans that change the cue structure rather than the price. Bamberg et al. (2011) and Graham-Rowe et al. (2011) review the evidence for such soft measures and find effects that are modest individually but meaningful in combination and cheap relative to civil works.

Recent Vietnamese evaluations provide timely settings in which to test that package rather than assume transferability from car-dominated countries. Work on fare-free public transport for older adults examines a population for whom motorcycle dependence and declining ability to ride intersect (Nguyen et al., 2026b), while research on electric motorcycle ride-hailing tests whether technological decarbonisation changes the mobility regime or reproduces it in a cleaner form (Nguyen et al., 2026a). The relevant policy comparison is therefore not rail versus motorcycles in isolation, but combinations of pricing, feeder reliability, shared mobility, parking and personalised support evaluated for distinct life-course and household contexts.

Two corridor-specific additions follow from our data. First, the 11.1% who name infrastructure include 22 respondents whose specific complaint is the absence of secure parking for a personal vehicle at stops — and this group lives closest to the stations (median 882 m). Secure motorcycle parking at station forecourts is a small, cheap intervention aimed at a precisely identified and

spatially concentrated group, and it works with the incumbent regime rather than against it. Second, the enterprise findings imply that any kerbside reorganisation along the corridor should be accompanied by a transition instrument for affected micro-enterprises, since the population at risk is thin-margined, short-tenured and has no evident alternative location.

7 Limitations

Five limitations bound the claims made here, and the first two are serious.

- *OpenStreetMap completeness confounds the before–after comparison.* Mapped transit objects in the study window grew by 160% between snapshots, and the composition of that growth — platform objects rising from 677 to 2183 while highway=bus_stop objects fell from 128 to 24 — indicates that much of it is retagging and route-relation mapping rather than construction. We audited rather than assumed, reported the audit before any before–after result, verified the fourteen metro stations individually, and confine the snapshot comparison to trend statements. But the accessibility gain in Table 6 and Figure 5 cannot be attributed to physical provision, and no analysis in this paper should be read as measuring infrastructure delivery.
- *No behavioural outcome after opening.* The survey predates commercial operation, so this paper measures a change in opportunity and a baseline of attribution, not a change in behaviour. It establishes who stood to benefit from the investment, independently of whether anyone changed what they do. A repeat wave on the 190 residents who consented to further contact would convert this into a genuine before–after design and is the single highest-value extension available.
- *Self-reported, single-select barriers.* Q2.8 permits one answer, so the instrument recovers the dominant perceived barrier rather than the full set, and self-reports may rationalise. The accessibility tests in Sections 5.4 and 5.5 are our response to the second concern: they cross-validate the reports against measurement, which is rarely done in the barriers literature. Kroesen et al. (2017) caution that attitudes and behaviour are mutually causal, and we make no claim about the direction of causation between naming habit and practising it.
- *No service-quality data.* No official open GTFS feed exists for Ho Chi Minh City. OpenStreetMap yields stop locations but not scheduled or realised headways, missed connections, excess waiting time or day-to-day reliability, so this is an analysis of spatial accessibility rather than service quality. Waiting and interchange generally carry greater perceived disutility than an equivalent amount of in-vehicle time (Wardman, 2004; Iseki and Taylor, 2009). Officials who name journey time may therefore be responding to unreliable feeder service as much as to vehicle speed. We cannot test that interpretation, and the diagnostic-gap result establishes only that the two groups explain non-use differently, not that either account is factually wrong. A follow-up should combine automatic vehicle-location records or repeated stop observations with passenger-reported waiting and missed-transfer experience.
- *Small samples for two strands.* The enterprise ($n = 80$) and officials ($n = 23$) samples support description and non-parametric comparison only. The enterprise sample cannot represent the full sectoral, spatial and tenure diversity of the economic ecosystem along Metro Line 1, estimate the prevalence of vulnerability, or identify a causal effect of station proximity or kerbside reorganisation. All enterprise analysis is therefore exploratory: the exposure-band result is one significant test among five, and the weakly separated clusters are profiles rather than

population types. Future research should purposively sample enterprises by sector, distance, tenure and dependence on kerbside stopping, then combine repeated customer/revenue measures with in-depth interviews of owners and workers. Interviews conducted before and after sidewalk clearance, access restrictions or relocation would be especially valuable for identifying the mechanisms linking pedestrian-access improvements to changes in deliveries, passing trade, operating costs and livelihood security. The resident sample ($n = 460$), not the enterprise strand, carries the paper's main quantitative claims.

8 Conclusion

This paper reports the behavioural state of a motorcycle-dependent corridor in Ho Chi Minh City in the last window before its first metro line began carrying passengers. In that corridor 83.5% of residents rode a private motorcycle and 1.7% took the bus. Asked why they did not use public transport, 68% named habit and 11.1% named an infrastructural deficit.

The contribution is not that ratio but its validation. Using two dated OpenStreetMap snapshots we measured, for every surveyed dwelling, the straight-line distance to the nearest stop, the walking distance along the pedestrian network, and the detour between them — and found that the barrier a resident names bears no relation to the conditions that resident faces. The group whose stated barrier was most plausibly physical, the 71 who said walking to stops was uncomfortable, walk no further and no less directly than anyone else. On that basis the perceptual share of the barrier structure is 83.4% against an infrastructural 11.1%. Motorcycle dependence is meanwhile unpredictable from the built environment and rises with income, which removes affordability as an explanation and leaves lock-in.

Two further results give the finding its policy edge. The corridor's micro-enterprises — almost all with five staff or fewer, almost all consumer-facing, half of them groceries or food service — depend on precisely the kerbside stopping behaviour a successful mode shift would remove, so the transition carries a livelihood cost that transport appraisal does not currently price. And the officials responsible for the corridor diagnose the problem as journey time, 51 percentage points away from the population's own account, which means the instrument most likely to be selected is not aimed at the barrier most often reported.

The recommendation follows without requiring anyone to stop building. Sustain the infrastructure programme, and add to it a behavioural component sized to the share of the barrier structure that infrastructure cannot reach, together with secure station parking for the small, spatially concentrated group whose complaint is exactly that, and a transition instrument for the micro-enterprises whose trade depends on the incumbent regime. Above all, repeat this survey. The baseline reported here cannot be collected again, and its value is realised only when there is a second wave to compare it with.

Data and code availability

All open data used here are freely redistributable: Geofabrik OpenStreetMap extracts for Viet Nam dated 1 January 2023 and 1 January 2026 (© OpenStreetMap contributors, ODbL); GHS-

POP R2023A and GHS-BUILT-S R2023A (European Commission Joint Research Centre, CC BY 4.0); Sentinel-2 derived built-up indices (Copernicus); and the Provincial Competitiveness Index 2021–2025 (VCCI). The complete analysis — OpenStreetMap extraction, accessibility computation, every statistical test, every table and every figure in this paper — is contained in a single annotated Jupyter notebook (Paper3_Analysis.ipynb) that runs end to end from the raw files and writes results.json, from which every quantity quoted in this text is drawn. Survey microdata contain household coordinates and are available from the corresponding author under a data-use agreement; a coordinate-masked version is available without restriction.

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